

ACmK-OT Clinical Operations Platform

Board of Nursing Compliance, System Architecture, and Clinical Safety Documentation

This document serves as a formal, regulator-facing description of the ACmK-OT Clinical Operations Platform. It details system architecture, clinical accountability mechanisms, crisis response controls, auditability, and interoperability in language aligned with U.S. State Boards of Nursing expectations.

1. System Purpose and Clinical Scope

ACmK-OT is a clinical operations engine designed to support licensed nursing staff in real-time patient care while enforcing individual accountability, scope-of-practice boundaries, and documentation integrity. The system prioritizes patient safety, nurse decision support, and defensible clinical workflows.

- Real-time nurse-to-patient assignment enforcement
- Shift-based authority and authentication
- Crisis escalation and staffing control
- Automatic documentation as a byproduct of care actions

2. Role-Based Accountability Model

ACmK-OT enforces role-specific access and authority through a RoleSession engine. Each clinical action is explicitly bound to a verified individual acting within their licensed role during an active shift.

- Nurse: Direct patient care, intervention acceptance, documentation
- Charge Nurse: Unit oversight, staffing coordination, crisis management
- Physician: Orders, escalation, medical decision authority
- Admin/Audit: Read-only compliance and forensic review

3. Assignment Engine and Continuity of Care

Patient assignments are managed through a deterministic Assignment Engine that ensures clear responsibility for every patient at all times. Assignments update automatically in response to staffing changes or crisis events.

- No unassigned patients during normal or crisis operations
- Temporary reassignment during emergencies
- Automatic restoration post-crisis

4. Crisis Coordination and Emergency Response

The Crisis Coordinator manages inbound ambulance alerts and internal emergencies by controlling staffing acknowledgements and maintaining patient coverage.

- Real-time availability acknowledgements
- Coverage thresholds enforced before escalation
- Prevention of unsafe nurse over-allocation

5. Audit, Documentation, and Legal Defensibility

ACmK-OT maintains an immutable audit ledger of all clinical actions. This ledger supports internal quality review, legal discovery, and Board of Nursing investigations.

- Actor identity and role at time of action
- Timestamped intervention records
- Authorization method recorded for every action

6. FHIR Interoperability and Data Exchange

ACmK-OT supports optional FHIR R4 interoperability through a translation adapter. This ensures compatibility with existing EMR ecosystems without compromising system integrity.

7. Board of Nursing Alignment Statement

The ACmK-OT platform enforces principles consistent with State Boards of Nursing, including individual accountability, scope-of-practice enforcement, supervision, and defensible documentation.

This system is designed to support—not replace—clinical judgment while ensuring that all actions are attributable, reviewable, and compliant with professional nursing standards.